Stokes' Theorem

Monday, June 9, 2025 5:58 AM

20 Transformation Between Surface and Line Integrals

20.1 We have already seen a version of Green's Theorem written as:

$$\iint_R (\vec{\nabla} \times \vec{F}) \cdot \hat{k} \, dx dy = \oint_C \vec{F} \cdot d\vec{r}$$

where $\vec{F} = F_1 \hat{\imath} + F_2 \hat{\jmath}$ is embedded as $\vec{F} = \vec{F}_1 \hat{\imath} + F_2 \hat{\jmath} + 0 \hat{k}$

There is nothing special about the xy-plane.

e.g. $\vec{G} = G_2 \hat{\jmath} + G_3 \hat{k}$ can be embedded as $\vec{G} = 0\hat{\imath} + G_2 \hat{\jmath} + G_3 \hat{k}$ and we get:

$$\iint_{R'} (\vec{\nabla} \times \vec{G}) \cdot \hat{\imath} \, dy dz = \oint_{C'} \vec{G} \cdot d\vec{r}$$

e.g. $\vec{H} = H_1 \hat{\imath} + H_3 \hat{k}$ embedded as $\vec{H} = H \hat{\imath} + 0 \hat{\jmath} + H_3 \hat{k}$ gives

$$\iint_{\bar{R}} (\vec{\nabla} \times \vec{H}) \cdot \hat{\jmath} \, dx dz = \oint_{\bar{C}} \vec{H} \cdot d\vec{r}$$

20.2

THEOREM 1

Stokes's Theorem[9]
(Transformation Between Surface and Line Integrals)

Let S be a piecewise smooth[9] oriented surface in space and let the boundary of S be a piecewise smooth simple closed curve C. Let $\mathbf{F}(x, y, z)$ *be a continuous vector function that has continuous first partial derivatives in a domain in space containing S. Then*

$$\iint_S (\text{curl } \mathbf{F}) \cdot \mathbf{n} \, dA = \oint_C \mathbf{F} \cdot \mathbf{r}'(s) \, ds. \tag{2}$$

Here $\mathbf{n}$ *is a unit normal vector of S and, depending on* $\mathbf{n}$*, the integration around C is taken in the sense shown in Fig. 254. Furthermore,* $\mathbf{r}' = d\mathbf{r}/ds$ *is the unit tangent vector and s the arc length of C.*

In components, formula (2) becomes

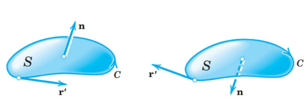


Fig. 254. Stokes's theorem

$$\iint_R \left[\left(\frac{\partial F_3}{\partial y} - \frac{\partial F_2}{\partial z} \right) N_1 + \left(\frac{\partial F_1}{\partial z} - \frac{\partial F_3}{\partial x} \right) N_2 + \left(\frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} \right) N_3 \right] du \, dv \tag{2*}$$
$$= \oint_{\bar{C}} (F_1 \, dx + F_2 \, dy + F_3 \, dz).$$

(2*)

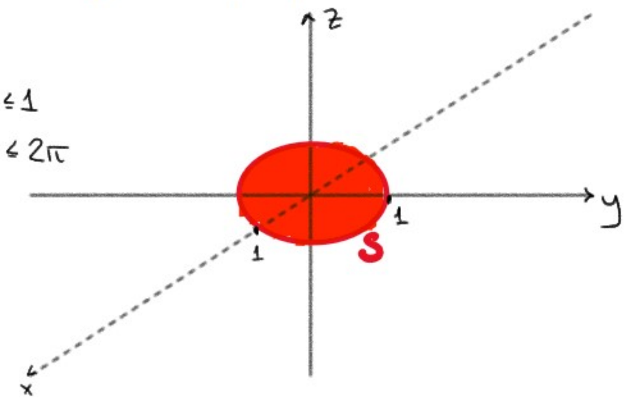
$$= \oint_{\overline{C}} (F_1\, dx + F_2\, dy + F_3\, dz).$$

Here, $\mathbf{F} = [F_1,\ F_2,\ F_3]$, $\mathbf{N} = [N_1,\ N_2,\ N_3]$, $\mathbf{n}\, dA = \mathbf{N}\, du\, dv$, $\mathbf{r}'\, ds = [dx,\ dy,\ dz]$, *and* R *is the region with boundary curve* $\overline{C}$ *in the uv-plane corresponding to* S *represented by* $\mathbf{r}(u, v)$.

20.3 Example.

Verify Stokes' Theorem for:

$$\mathbf{F} = [y^3, -x^3, 0], \quad S\colon x^2 + y^2 \leqq 1, \quad z = 0$$



Parameterize S:

$x(u,v) = u\cos v$, $\quad 0 \le u \le 1$

$y(u,v) = u\sin v$, $\quad 0 \le v \le 2\pi$

$z(u,v) = 0$

Then $\vec{r}_u = \begin{bmatrix} \cos v \\ \sin v \\ 0 \end{bmatrix}$ while $\vec{r}_v = \begin{bmatrix} -u\sin v \\ u\cos v \\ 0 \end{bmatrix}$

So

$$\vec{N} = \vec{r}_u \times \vec{r}_v = \begin{bmatrix} \cos v \\ \sin v \\ 0 \end{bmatrix} \times \begin{bmatrix} -u\sin v \\ u\cos v \\ 0 \end{bmatrix}$$

$$= \hat{k}\left(u\cos^2 v - (-u\sin^2 v)\right) = u\hat{k} = \begin{bmatrix} 0 \\ 0 \\ u \end{bmatrix}$$

Now:

$$\vec{\nabla} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ y^3 & -x^3 & 0 \end{vmatrix} = \hat{k}\left(\frac{\partial}{\partial x}(-x^3) - \frac{\partial}{\partial y}(y^3)\right)$$

$$= \hat{k}\left(-3x^2 - 3y^2\right) = (-3)\begin{bmatrix} 0 \\ 0 \\ x^2+y^2 \end{bmatrix}$$

Using $\left.\begin{array}{l} x = u\cos v \\ y = u\sin v \end{array}\right\} x^2+y^2 = u^2$

So on the surface S: $\vec{\nabla} \times \vec{F} = -3u^2\hat{k}$

Then

$$\iint_S (\vec{\nabla}\times\vec{F})\cdot\hat{n}\, dA = \iint_S (\vec{\nabla}\times\vec{F})\cdot\vec{N}(u,v)\, dv\, du$$

$$= \int_0^1 \int_0^{2\pi} (-3u^2\hat{k})\cdot(u\hat{k})\, dv\, du$$

$$= \int_0^1\int_0^{2\pi} -3u^3\, dv\, du$$

$$= \int^1 (-3)(2\pi)\, u^3\, du$$

$$= \int_0^1 (-3)(2\pi)\,u^3\,du$$

$$= -6\pi\left[\frac{u^4}{4}\right]_0^1 = \frac{-6\pi}{4}$$

$$= \frac{-3\pi}{2}$$

On the other hand,

boundary of S is the unit circle $C: x^2+y^2=1,\ z=0$

parameterized as going counter-clockwise

$$\vec{r}(t) = \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix} = \begin{bmatrix} \cos(t) \\ \sin(t) \\ 0 \end{bmatrix}. \qquad 0 \le t \le 2\pi$$

[Note: not arclength parameterization]

Thus

$$\vec{r}\,'(t) = \begin{bmatrix} -\sin(t) \\ \cos(t) \\ 0 \end{bmatrix}$$

and along C: $\vec{F} = \begin{bmatrix} y^3 \\ -x^3 \\ 0 \end{bmatrix} = \begin{bmatrix} \sin^3(t) \\ -\cos^3(t) \\ 0 \end{bmatrix}$

Therefore

$$\oint_C \vec{F}\cdot d\vec{r} = \int_0^{2\pi} \begin{bmatrix} \sin^3(t) \\ -\cos^3(t) \\ 0 \end{bmatrix} \cdot \begin{bmatrix} -\sin(t) \\ \cos(t) \\ 0 \end{bmatrix} dt$$

$$= \int_0^{2\pi} \left(-\sin^3(t)\sin(t) - \cos^3(t)\cos(t) + 0\right) dt$$

$$= -\int_0^{2\pi} \left(\sin^4(t) + \cos^4(t)\right) dt$$

$$= -\int_0^{2\pi} \left(\sin^4(t) + 2\sin^2 t\cos^2 t + \cos^4(t) - 2\sin^2 t\cos^2 t\right) dt$$

$$= -\int_0^{2\pi} \left[\left(\sin^2 t + \cos^2 t\right)^2 - 2\sin^2 t\cos^2 t\right] dt$$

$$= -\int_0^{2\pi} \left[1 - 2\sin^2 t\cos^2 t\right] dt$$

$$= -\int_0^{2\pi} \left(1 - \frac{1}{2}\sin^2(2t)\right) dt$$

$$= -\int_0^{2\pi} \left[\frac{3}{4} + \frac{1}{4} - \frac{1}{2}\sin^2(2t)\right] dt$$

$$= -\int_0^{2\pi} \left(\frac{3}{4} + \frac{1 - 2\sin^2(2t)}{4}\right) dt$$

$$= -\int_0^{2\pi} \left(\frac{3}{4} + \frac{1}{4}\cos(4t)\right) dt$$

$$= -\int_0 \left(\frac{3}{4} + \frac{1}{4}\cos(4t)\right)dt$$

$$= -\int_0^{2\pi} \frac{3}{4}dt - \frac{1}{4}\int_0^{2\pi} \cos(4t)\,dt$$

$$= -\frac{3}{4}(2\pi) - \frac{1}{4}\left[\frac{1}{4}\sin(4t)\right]_0^{2\pi}$$

$$= -\frac{3\pi}{2} - 0 = -\frac{3\pi}{2}$$

We get the same answer from $\iint_S (\vec{\nabla}\times\vec{F})\cdot\hat{n}\,dA$ and $\oint_C \vec{F}\cdot d\vec{r}$, verifying Stokes' Theorem.

20.4

EXAMPLE 1 Verification of Stokes's Theorem

Before we prove Stokes's theorem, let us first get used to it by verifying it for $\mathbf{F} = [y, z, x]$ and S the paraboloid (Fig. 255)

$$z = f(x, y) = 1 - (x^2 + y^2), \qquad z \geqq 0.$$

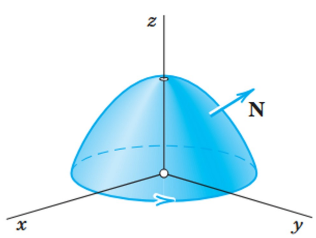


Fig. 255. Surface S in Example 1

Solution. The curve C, oriented as in Fig. 255, is the circle $\mathbf{r}(s) = [\cos s, \sin s, 0]$. Its unit tangent vector is $\mathbf{r}'(s) = [-\sin s, \cos s, 0]$. The function $\mathbf{F} = [y, z, x]$ on C is $\mathbf{F}(\mathbf{r}(s)) = [\sin s, 0, \cos s]$. Hence

$$\oint_C \mathbf{F} \bullet d\mathbf{r} = \int_0^{2\pi} \mathbf{F}(\mathbf{r}(s)) \bullet \mathbf{r}'(s)\,ds = \int_0^{2\pi} [(\sin s)(-\sin s) + 0 + 0]\,ds = -\pi.$$

We now consider the surface integral. We have $F_1 = y$, $F_2 = z$, $F_3 = x$, so that in (2*) we obtain

$$\operatorname{curl} \mathbf{F} = \operatorname{curl}[F_1, \quad F_2, \quad F_3] = \operatorname{curl}[y, \quad z, \quad x] = [-1, \quad -1, \quad -1].$$

A normal vector of S is $\mathbf{N} = \operatorname{grad}(z - f(x, y)) = [2x, 2y, 1]$. Hence $(\operatorname{curl} \mathbf{F}) \bullet \mathbf{N} = -2x - 2y - 1$. Now $\mathbf{n}\,dA = \mathbf{N}\,dx\,dy$ (see (3*) in Sec. 10.6 with x, y instead of u, v). Using polar coordinates r, θ defined by $x = r\cos\theta$, $y = r\sin\theta$ and denoting the projection of S into the xy-plane by R, we thus obtain

$$\iint_S (\operatorname{curl} \mathbf{F}) \bullet \mathbf{n}\,dA = \iint_R (\operatorname{curl} \mathbf{F}) \bullet \mathbf{N}\,dx\,dy = \iint_R (-2x - 2y - 1)\,dx\,dy$$

$$= \int_{\theta=0}^{2\pi} \int_{r=0}^{1} (-2r(\cos\theta + \sin\theta) - 1)r\,dr\,d\theta$$

$$= \int_{\theta=0}^{2\pi} \left(-\frac{2}{3}(\cos\theta + \sin\theta) - \frac{1}{2}\right)d\theta = 0 + 0 - \frac{1}{2}(2\pi) = -\pi. \quad \blacksquare$$

20.5

EXAMPLE 3 Evaluation of a Line Integral by Stokes's Theorem

Evaluate $\int_C \mathbf{F} \bullet \mathbf{r}'\,ds$, where C is the circle $x^2 + y^2 = 4$, $z = -3$, oriented counterclockwise as seen by a person standing at the origin, and, with respect to right-handed Cartesian coordinates,

$$\mathbf{F} = [y, \quad xz^3, \quad -zy^3] = y\mathbf{i} + xz^3\mathbf{j} - zy^3\mathbf{k}.$$

Solution. As a surface S bounded by C we can take the plane circular disk $x^2 + y^2 \leqq 4$ in the plane $z = -3$. Then $\mathbf{n}$ in Stokes's theorem points in the positive z-direction; thus $\mathbf{n} = \mathbf{k}$. Hence $(\operatorname{curl} \mathbf{F}) \bullet \mathbf{n}$ is simply the component of $\operatorname{curl} \mathbf{F}$ in the positive z-direction. Since $\mathbf{F}$ with $z = -3$ has the components $F_1 = y$, $F_2 = -27x$, $F_3 = 3y^3$, we thus obtain

$$(\operatorname{curl} \mathbf{F}) \bullet \mathbf{n} = \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = -27 - 1 = -28.$$

20·5

EXAMPLE 3 Evaluation of a Line Integral by Stokes's Theorem

Evaluate $\int_C \mathbf{F} \cdot \mathbf{r}' ds$, where C is the circle $x^2 + y^2 = 4$, $z = -3$, oriented counterclockwise as seen by a person standing at the origin, and, with respect to right-handed Cartesian coordinates,

$$\mathbf{F} = [y, \quad xz^3, \quad -zy^3] = y\mathbf{i} + xz^3\mathbf{j} - zy^3\mathbf{k}.$$

Solution. As a surface S bounded by C we can take the plane circular disk $x^2 + y^2 \leqq 4$ in the plane $z = -3$. Then $\mathbf{n}$ in Stokes's theorem points in the positive z-direction; thus $\mathbf{n} = \mathbf{k}$. Hence $(\text{curl } \mathbf{F}) \cdot \mathbf{n}$ is simply the component of curl $\mathbf{F}$ in the positive z-direction. Since $\mathbf{F}$ with $z = -3$ has the components $F_1 = y$, $F_2 = -27x$, $F_3 = 3y^3$, we thus obtain

$$(\text{curl } \mathbf{F}) \cdot \mathbf{n} = \frac{\partial F_2}{\partial x} - \frac{\partial F_1}{\partial y} = -27 - 1 = -28.$$

Hence the integral over S in Stokes's theorem equals -28 times the area 4π of the disk S. This yields the answer $-28 \cdot 4\pi = -112\pi \approx -352$. Confirm this by direct calculation, which involves somewhat more work. ■

20·6 <u>Work-done around a closed curve</u>:

For a force field $\vec{F}$, we define work along a path C to be $\int_C \vec{F} \cdot d\vec{r}$

If C is a closed loop, then by Stokes' Theorem:

$$\text{Work-done} = \oint_C \vec{F} \cdot d\vec{r} = \iint_S (\vec{\nabla} \times \vec{F}) \cdot \hat{n}\, dA,$$

where S is <u>any</u> surface with C as the boundary in the correct orientation

When $\vec{F}$ is conservative, i.e., $\vec{F} = \vec{\nabla} f$, then $\vec{\nabla} \times \vec{F} = \vec{0}$.

Can we say that the work-done is also zero?

- only guaranteed if there is a <u>simply connected domain D</u> of $\vec{F}$ that contains S and C.

In a simply connected domain, work-done by a <u>conservative</u> force on a closed loop will <u>always</u> be zero.